

CO 4		✓		✓
CO 5	✓	✓		✓
CO 6	✓			

Programme	BSc Statistics				
Course Code	STA3MN208 (P)				
Course Title	Statistics for critical thinking III				
Type of Course	Minor				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic understanding of introductory statistical concepts. Familiarity				

	with the fundamentals of probability.
Course Summary	This course examines different ways to analyse data to make meaningful conclusions about the larger population from it is drawn. Course also explores ways to describe relationships between different variables in a data matrix.

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain the concept of a sampling distribution and calculate standard error for different statistics.	U	C	Instructor-created exams / Quiz
CO2	Explain central limit theorem and its importance in statistics.	U	C	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Construct and interpret confidence intervals for proportions, including understanding the margin of error and changing the confidence level and analyze data to help entrepreneurial decisions using critical thinking skills.	U	C	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Analyze the difference of two proportions using hypothesis tests and confidence intervals.	U	C	Instructor-created exams / Home Assignments
CO5	Perform paired data analysis using paired t-tests and interpret the results and critically evaluate ethical implications of statistical methods aligning with human values.	U	f	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Conduct ANOVA tests to compare means across multiple groups and interpret ANOVA tables using R software.	Ap	P	Viva Voce/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)				

- Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)
--

COURSE CONTENT

Module		Content	Hours (45+ 30)	Marks (70)
		Statistical Inference	12	17
1	1	Point estimates and sampling variability, Sampling error, Bias.	1	
	2	Sampling distribution, standard error, Central Limit Theorem.	1	
	3	Applying the Central Limit Theorem to a real-world setting, More details regarding the Central Limit Theorem.	2	
	4	Confidence intervals for a proportion, Capturing the population parameter.	2	
	5	Constructing a 95% confidence interval	1	
	6	Changing the confidence level, margin of error, case studies, Interpreting confidence intervals.	1	
	7	Hypothesis testing for a proportion, null hypothesis and alternative hypothesis, Type I and Type II errors, Formal testing using p-values.	4	
		Sections from References: Unit 1-7: 5.1,5.2,5.3 [Ref 2]		
		Hypothesis testing	12	17
2	8	Inference for categorical data, Inference for a single proportion, Confidence intervals for a proportion.	2	
	9	Hypothesis testing for a proportion, Choosing a sample size when estimating a proportion.	1	
	10	Difference of two proportions, Sampling distribution of the difference of two proportions.	2	
	11	Hypothesis tests for the difference of two proportions.	1	
	12	Testing for goodness of fit using chi-square: Creating a test statistic for one-way tables, The chi-square test statistic.	2	
	13	The chi-square distribution and finding areas,	2	

		Finding a p-value for a chi-square distribution, Evaluating goodness of fit for a distribution.		
	14	Testing for independence in two-way tables, The chi-square test for two-way tables.	2	
		Sections from References: Unit 8-14: 6.1-6.4 [Ref 2]		
		Small sample tests	14	19
3	15	Inference for numerical data: One-sample means with the t-distribution, The sampling distribution of sample mean, Introducing the t-distribution, One sample t-tests.	3	
	16	Paired data, paired t-test. Difference of two means,	2	
	17	Hypothesis tests for the difference of two means, Confidence interval for a difference of means	4	
	18	Comparing many means with ANOVA: Core ideas of ANOVA, Analysis of variance (ANOVA) and the F -test.	3	
	19	Reading an ANOVA table from software, Multiple comparisons and controlling Type 1 Error rate.	2	
		Sections from References: Unit 15-19: 7.1-7.5 [Ref 2]		
		Regression	7	17
4	20	Introduction to linear regression: Fitting a line, residuals, and correlation, Describing linear relationships with correlation.	3	
	21	Least squares regression, Conditions for the least squares line, Finding the least squares line.	2	
	22	Interpreting regression model parameter estimates, Using R ² to describe the strength of a fit, Categorical predictors with two levels.	2	
		Sections from References: Unit 20-24:8.1-8.4 [Ref 2]		
5	PRACTICUM		30	
	Do practice problems in R software from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Test Concerning Means-One sample		
	2	Analytical Methods of checking assumption of normality of parent population		

	3	Test of Significance for difference of two population means		
	4	Test of Significance for difference of two population proportions		
	5	ANOVA		
	6	Correlation		
	7	Inference procedures for correlation coefficient		
	8	Linear regression		
	Sections from References: Unit 1: 4.5[Ref 5] Unit 2:4.5[Ref 5] Unit 3: 4.5[Ref 5] Unit 4: 4.5[Ref 5] Unit 5: 4.5[Ref 5] Unit 6: 5.2[Ref 5] Unit 7: 5.3[Ref 5] Unit 8: 5.4[Ref 5]			
	Books and References: 1. Asthana, H. S., & Bhushan, B. (2016). <i>Statistics for social sciences (with SPSS applications)</i> . PHI Learning Pvt. Ltd.. 2. Diez, D. M., Barr, C. D., & Cetinkaya-Rundel, M. (2019). <i>OpenIntro statistics</i> . Boston, MA, USA:: OpenIntro. 3. Aron, A., Coups, E. J., & Aron, E. N. (2013). <i>Statistics for the behavioral and social sciences: A brief course: Pearson new international edition</i> . Pearson Higher Ed. 4. Sirkin, R. M. (2006). <i>Statistics for the social sciences</i> . Sage. 5. Sudha G Purohith, Sharad D Core, Shailaja R Deshmukh ,Statistics Using R(2015). 6. Mukherjee, S. P., Sinha, B. K., & Chattopadhyay, A. K. (2018). <i>Statistical methods in social science research</i> (pp. 29-37). Springer Singapore. 7. Gupta, S. C. and Kapoor, V. K. (2002). <i>Fundamentals of Mathematical Statistics</i> . , 11 th edition, Sulthan Chand, New Delhi. 8. Gupta, S. C. and Kapoor, V. K. (2007). <i>Fundamentals of applied Statistics</i> , Sultan Chand and Sons.			

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	-	-	-	-	2	-	-	-	3	-	-	-
CO 2	-	-	-	-	-	-	3	-	-	-	-	-
CO 3	2	2	-	-	-	-	1	-	-	-	3	-
CO 4	-	-	2	3	1	3	-	3	2	-	-	-
CO 5	3	-	-	-	-	-	-	-	-	-	-	3
CO 6	-	-	3	-	-	2	-	-	-	3	3	-

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :